$x^2 - y^2 - z^2 = 0$

Let $\vec{x} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$

⇒ Quadratic form

$\vec{x}^T M \vec{x} = 0$

$$\begin{pmatrix} x & y & z \end{pmatrix} \begin{pmatrix} a & h & g \\ h & b & f \\ g & f & c \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0$$

⇒ $ax^2 + by^2 + cz^2 + 2hxy + 2gxz + 2fyz = 0$

⇒ $M = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$

$$\nabla(\vec{x}^T M \vec{x}) = 2M\vec{x}$$

∴ Normal vector at point $\vec{x}_0$ is

$\vec{N} = M\vec{x}_0$

∴ $\vec{x}_0 = \begin{bmatrix} 1 \\ 0 \\ \sqrt{2} \end{bmatrix}$

⇒ $\vec{N} = \begin{bmatrix} 1 \\ 0 \\ -\sqrt{2} \end{bmatrix}$

⇒ Unit vector in direction of $\vec{N}$ is

$$\frac{\vec{N}}{\|\vec{N}\|} = \frac{1}{\sqrt{3}} \begin{bmatrix} \sqrt{2} \\ 0 \\ -1 \end{bmatrix}$$

$$2x^2 - y^2 - z^2 = 0$$

Let $\vec{x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$

$\Rightarrow$ Quadratic Form:

$$\vec{x}^T M \vec{x} = 0$$

$$\begin{bmatrix} x & y & z \end{bmatrix} \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$$

$$\Rightarrow \quad ax^2 + by^2 + cz^2 + 2dxy + 2exz + 2fyz = 0$$

$$\Rightarrow M = \begin{bmatrix} 2 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

$$\nabla\left(\vec{x}^T M \vec{x}\right) = 2M\vec{x}$$

$\therefore$ Normal vector at point $\vec{x_0}$ is

$$\vec{N} = M\vec{x_0}$$

$$\therefore \vec{x}_0 = \begin{bmatrix} 1 \\ 0 \\ \sqrt{2} \end{bmatrix}$$

$$\Rightarrow \vec{N} = \begin{bmatrix} 2 \\ 0 \\ -\sqrt{2} \end{bmatrix}$$

$\Rightarrow$ Unit vector in direction of $\vec{N}$ is

$$\frac{\vec{N}}{\|\vec{N}\|} = \frac{1}{\sqrt{3}} \begin{bmatrix} \sqrt{2} \\ 0 \\ -1 \end{bmatrix}$$